

EDUCATION

PhD – Atmospheric Sciences Seattle, WA University of Washington	Defended January 2025
<i>Dissertation: Convective Upscale Growth in Central Argentina: Environmental Conditions and the Role of the South American Low-level Jet</i>	
MS – Atmospheric Sciences Seattle, WA University of Washington	2021
<i>Thesis: New Insights into the South American Low-Level Jet from RELAMPAGO Observations</i>	
BS - Atmospheric Science, BS - Environmental Engineering Ithaca, NY Cornell University	2018
<i>Honors Thesis: Exploring the Impacts of Energy Generation Composition on the Future Climate</i>	

EXPERIENCE

Research Assistant Seattle, WA <i>University of Washington, Atmospheric Sciences</i>	September 2018-Present
- Investigated how variations in the thermodynamic and dynamic environment influence convective storm organization and characteristics using observations and high-resolution convective simulations - Analyzed large meteorological datasets, identifying spatial/temporal patterns and creating geospatial visualizations using Python packages and GIS - Collaborated with scientists across agencies to improving modeling of convective processes - Published and presented findings at conferences and public meetings	
Research Assistant Ithaca, NY <i>Cornell University, Atmospheric Sciences</i>	August 2016-May 2018
- Created simple climate models to simulate radiative forcing and the corresponding global temperature changes incorporating a variety of economic and emissions scenarios based upon projections of electricity generation makeup and electricity demand - Managed long-term multi-file projects on unix servers	
Student Intern Charleston, WV <i>National Weather Service</i>	July 2017-August 2017
- Assisted with briefing state and local emergency managers and provided support during real-time weather watches - Refined short- and long-range forecasts using remote meteorological and hydrological observational tools such as Doppler radar, satellite imagery, surface and upper air observations, and numerical models - Developed shell scripts to auto-generate specific graphics to aid in decision-support - Conducted E-19 flood stage and post-storm surveys - Managed surface weather stations (rain gauges, temperature and humidity sensors)	
Assistant Analyst Whiteley, UK <i>National Air Traffic Services</i>	May 2017-June 2017
- Performed cost-benefit analysis of future airspace management projects by modeling incurred delay using NEST (Network Strategic Tool) to model future scenarios and Excel to sort data and present analysis - Assisted in the development of indices to forecast meteorological impacts at airports using METAR/TAF data and with machine learning - Researched and wrote report suggesting approaches for forecasting en-route weather delay based on forecastable meteorological parameters - Wrote report on radiative forcing used in training the Sustainable Aviation Group	
Research Assistant Ithaca, NY <i>Cornell University, Biological & Environ Engineering</i>	January 2016-May 2016
Researched the effect of Glyphosate (active pesticide ingredient) and Tallow Amine on the growth rate of <i>P. Putida</i> (a bacteria found in soil that is beneficial to plant growth)	

- Initiated and independently researched project assessing location specific weather alerts through analysis of temperature and precipitation data from the 30 largest cities in the US
- Analyzed conferences which represented potential business development opportunities

SELECTED PUBLICATIONS

Sasaki, C. R. S., A. K. Rowe, and L. A. McMurdie, 2025b: Initial Upscale Growth Environments in Central Argentina from a Convection-Permitting Simulation. *J. Geophys. Res. Atmos.* (in-prep).

Sasaki, C. R. S., A. K. Rowe, and L. A. McMurdie, 2025a: Environmental Conditions Leading to Observed Convective Organization in Central Argentina. *Mon. Wea. Rev.* (in-review).

Sasaki, C. R. S., A. K. Rowe, L. A. McMurdie, A. Varble, and Z. Zhang, 2024: Influences of the South American Low-Level Jet on the Convective Environment in Central Argentina Using a Convection-Permitting Simulation. *Mon. Wea. Rev.*, 152, 629–648, <https://doi.org/10.1175/MWR-D-23-0122.1>.

SELECTED PRESENTATIONS

Sasaki, C. R. S., A. K. Rowe, and L. McMurdie, 2024: Environmental Conditions Leading to Observed Convective Aggregation in Central Argentina. 21th Conf. on Mountain Meteorology, Amer. Meteor. Soc., <https://ams.confex.com/ams/21MOUNTAIN/meetingapp.cgi/Paper/444246>.

Sasaki, C. R. S., A. K. Rowe, and L. McMurdie, 2023: Low-Level Jet Influences on the Warm Season Convective Environment in Central Argentina. 20th Conf. on Mesoscale Processes, Amer. Meteor. Soc., <https://ams.confex.com/ams/WAFNWPMs/meetingapp.cgi/Paper/425171>.

AWARDS AND HONORS

Radar Observations of Clouds and Precipitation Summer School Travel Grant, Stony Brook University	2023
Graduate Student Distinguished Service Certificate, UW Atmospheric Sciences	2022
Achievement Rewards for College Scientists (ARCS) Fellowship	2018-2021
Distinction in research in Atmospheric Science, Cornell University	2018
Departmental Honors in Environmental Engineering, Cornell University	2018
Elected to membership in the Alpha Epsilon Honor Society	2017

COMMUNITY ENGAGEMENT

Cabinet Member , <i>King County Metro Fares Cabinet</i>	2024-Present
Board Member , <i>AMS STAC Weather Analysis and Forecasting Committee</i>	2023-Present
Peer Reviewer , https://www.webofscience.com/wos/author/record/3705759	2022-Present
Board Member , <i>Habitat for Humanity, Young Professionals Seattle-King County</i>	2018-2020, 2022-Present
Board Member , <i>Husky Experience Student Advisory Council</i>	2022-2023
Graduate Student Distinguished Visiting Lecture Organizer , <i>Department of Atmospheric Sciences, University of Washington</i>	2019-2023
Undergraduate Mentor , <i>Department of Atmospheric Sciences, University of Washington</i>	2019-2024

TEACHING EXPERIENCE

Instructor (Exploring Atmos. Sciences) Seattle, WA <i>University of Washington, Atmos. Sciences</i>	Spring 2021
Teaching Assistant (Weather) Seattle, WA <i>University of Washington, Atmos. Sciences</i>	Fall 2019

Technical Skills

Programming: Python, MATLAB, R, bash, Java, LATEX, IDL, HTML, CSS, GitHub
Visualizations: matplotlib, seaborn, cartopy, basemap, GrADS, Excel, AWIPS2, Photoshop, ArcGIS
Data Formats: netCDF, GRIB

Personal Interests

Hiking, sailing, WxChallenge forecasting, travelling, following the news